

Deep Learning for Panoramic Dental Radiographs: A Comprehensive Survey on Caries Detection, Periodontal Bone Loss Assessment and Explainable AI

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Abstract—Dental panoramic radiographs (OPGs) are the primary clinical tool for diagnosing caries and periodontal disease, yet manual interpretation is slow and varies substantially between examiners. Deep learning has advanced considerably, but most published systems address a single diagnostic task in isolation, leaving the clinical integration problem unsolved.

This paper surveys 47 deep learning studies published between 2021 and 2026, covering caries detection, severity staging, periodontal bone loss assessment, tooth segmentation, and explainable AI. Architectures examined include CNNs, YOLO-based detectors, U-Net variants, and transformer-based models. We formally define a unified multi-task dental AI framework that combines detection, localization, severity staging, quantification, and explainability within a single system.

Despite strong performance on individual sub-tasks, no existing approach delivers a complete, clinically usable pipeline. Five recurring gaps are identified: task fragmentation, absent severity staging, low explainability adoption, dataset bias, and limited clinical integration.

The analysis provides a structured foundation for building the next generation of clinically deployable dental AI systems.

Index Terms—Deep Learning, Dental Caries, Periodontal Disease, Panoramic Radiographs, Tooth Segmentation and Detection, Multi-Task Learning, Explainable AI

1. INTRODUCTION

Dental panoramic radiography, also known as OrthoPan-tomography (OPG), is the standard imaging tool in routine dental practice. A single panoramic exposure captures the full dentition, surrounding alveolar bone, and anatomical landmarks in one image, making it the primary modality for detecting dental caries, periodontal bone loss, impacted teeth, and other oral pathologies. Accurate interpretation nevertheless requires substantial expertise and is subject to inter-examiner variability, a problem that affects early-stage practitioners such as dental students and interns more acutely than experienced clinicians.

Dental caries, defined as the progressive bacterial demineralization of tooth enamel and dentine, affects approximately 2.3 billion people globally and is the most prevalent non-communicable disease worldwide [1]. Periodontal disease,

characterized by irreversible alveolar bone loss around the teeth, affects over 19% of the adult global population and is the leading cause of adult tooth loss [2]. Both conditions share a clinically consequential property: their early radiographic signs are subtle and easily missed. A faint radiolucency at the proximal surface of a premolar, or a 1–2 mm reduction in alveolar bone height, can alter the clinical decision yet remain visually ambiguous even for trained examiners.

Over the past five years, deep learning has produced notable gains in automated dental radiograph analysis. Convolutional neural networks (CNNs), YOLO-based object detectors, U-Net segmentation architectures, and transformer-based models have demonstrated competitive performance across individual diagnostic tasks such as caries detection, tooth segmentation, and periodontal bone loss estimation [3]–[5].

These advances have not, however, resolved the fragmentation that characterizes the field at the system level. Most published approaches target isolated tasks or single-disease objectives; the components required for a complete clinical workflow are rarely combined. The underlying problem is structural: models are tuned for task-specific metrics rather than the kind of holistic diagnostic reasoning that clinical use demands [6], [7].

Automated dental radiograph analysis is multi-dimensional by nature. A clinically viable system must handle multi-instance detection across all teeth, identify low-contrast early-stage lesions, measure anatomical landmarks such as the cemento-enamel junction (CEJ) and alveolar bone crest, and manage the co-occurrence of multiple diseases within a single panoramic image. This diagnostic process can be conceptualized as a pipeline, as illustrated in Fig. 1.

The absence of standardized datasets and evaluation protocols further limits reproducibility and cross-study comparability. A large proportion of studies rely on private, single-center data, which introduces dataset bias and reduces generalization to external clinical environments [7], [8]. Explainable AI (XAI) is also rarely implemented, despite its recognized role in

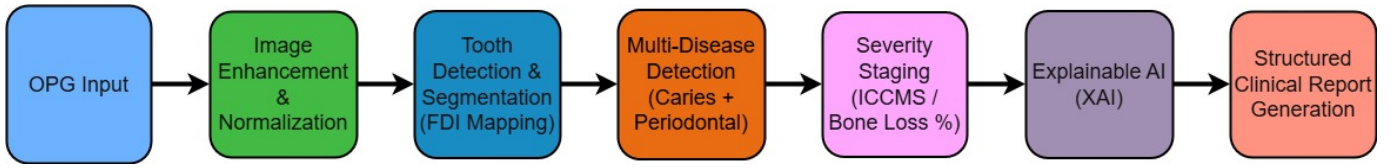


Fig. 1. System Pipeline of Proposed AI Model.

building clinical trust, supporting regulatory acceptance, and enabling interpretable output [9], [10]. Without explainability, combined with inconsistent validation practices, clinical deployment remains out of reach for most published systems.

Rather than surveying individual tasks or model architectures in isolation, this work takes a system-level perspective, examining how individual components might be combined into a unified diagnostic framework [11], [12].

The survey covers 47 papers published between 2021 and 2026, organized into six functional categories: caries detection and severity classification, periodontal disease and bone loss analysis, multi-task end-to-end diagnosis, explainable AI, dataset benchmarking and clinical validation, and tooth segmentation and numbering. The central finding is that no existing system simultaneously addresses caries detection, severity staging, periodontal bone loss quantification, and explainability within a single unified framework for panoramic radiographs.

The main contributions of this survey are as follows:

- A task-level taxonomy of deep learning systems for panoramic dental radiograph analysis, covering detection, segmentation, classification, and multi-task architectures.
- A formal problem formulation for unified multi-disease dental diagnosis, highlighting the integration requirements of detection, severity staging, quantification, and explainability.
- A cross-category comparative analysis of 47 studies, identifying inconsistencies in datasets, evaluation metrics, and validation protocols.
- A quantified gap analysis revealing key limitations in current research, including single-disease model focus, absence of automated severity staging, dataset fragmentation, and low adoption of explainable AI.
- A research roadmap toward clinically deployable, explainable, multi-task dental AI systems.

A unified, explainable, multi-task approach to dental AI is the logical next step toward systems that can be deployed in clinical practice, particularly for supporting junior clinicians and dental students.

2. SEARCH METHODOLOGY

This survey used a structured keyword search across five digital libraries: IEEE Xplore, PubMed/MEDLINE, Scopus, Google Scholar, and SpringerLink. Searches were conducted between January 2024 and March 2025, restricted to papers published from 2020 onward.

2.1 Search Strategy

Search queries were organized into five thematic clusters:

- *Caries detection on panoramic radiographs*: “dental caries” panoramic radiograph deep learning; “caries detection” panoramic radiograph CNN; “caries segmentation” orthopantomogram neural network.
- *Periodontal disease and bone loss*: “periodontal disease” panoramic radiograph deep learning; “alveolar bone loss” orthopantomogram AI; “CEJ” “bone crest” panoramic radiograph segmentation.
- *Multi-disease and end-to-end OPG systems*: “multi-task” dental panoramic radiograph deep learning; “dental disease detection” panoramic radiograph end-to-end.
- *Explainable AI in dental imaging*: “explainable AI” dental radiograph; “Grad-CAM” dental panoramic; “interpretable deep learning” dental imaging.
- *Dataset and clinical validation*: “dental panoramic radiograph dataset” deep learning; “OPG benchmark” AI; “clinical validation” dental AI panoramic.

2.2 Inclusion and Exclusion Criteria

Papers were included if they: (i) applied deep learning or machine learning to dental radiograph analysis; (ii) used panoramic, bitewing, or periapical radiographs as primary input; (iii) targeted detection, segmentation, classification, or staging of dental caries, periodontal disease, or related pathologies; and (iv) reported quantitative performance metrics. Papers were excluded if they focused exclusively on CBCT/3D imaging, addressed only non-dental imaging, or did not report reproducible quantitative results. Clinical reviews without AI models were included only when they provided foundational clinical definitions directly relevant to the survey scope.

2.3 Final Corpus

After applying the criteria, 47 papers were selected, spanning 2021–2026: 25 from 2025 (53%), 17 from 2024 (36%), 2 from 2026, 2 from 2023, and 1 from 2021. This distribution reflects the field’s rapid growth over recent years. Publication venues include BMC Oral Health (8 papers), IEEE (8), MDPI journals (9), Elsevier (4), Springer (3), and Scientific Reports/Nature Portfolio (3).

3. SYSTEM-LEVEL TAXONOMY AND PROBLEM FORMULATION

3.1 Formal Problem Definition

Panoramic dental radiograph analysis is a multi-component diagnostic problem that cannot be reduced to single-task

prediction. In clinical practice, meaningful diagnosis requires coordinated execution of several interdependent processes: localization of teeth, identification of pathological conditions, assessment of disease severity, and quantitative measurement of relevant anatomical parameters. Most deep learning systems in the reviewed literature address only one of these components [1], [6], [12], producing outputs that are diagnostically incomplete and difficult to use in a clinical setting.

A unified dental AI system maps an input OPG to structured diagnostic outputs through six core functional components: detection, localization, severity staging, quantification, explainability, and clinical integration. To our knowledge, no reviewed study in the corpus of 47 papers fully addresses all six components simultaneously on panoramic radiographs. That gap is the primary motivation for this survey.

These six core components are defined as follows:

- **Detection:** Identification of pathological conditions from radiographic features (carious lesions, periodontal bone loss, periapical pathology, impacted teeth). YOLO-family architectures dominate this task [3], [13]–[16].
- **Localization:** Spatial identification of affected teeth through bounding boxes or polygon segmentation. Tooth-level localization using FDI numbering is present in pipeline systems [4], [14]–[16] but absent in classification-only approaches [17]–[19].
- **Severity Staging:** Classification of disease progression along clinically meaningful scales. For caries: enamel, dentine, pulp involvement [15], [20]. For periodontal disease: Stages I–IV per 2018 AAP/EFP classification [21]. This is the most critically underaddressed component, with only two C1 papers providing caries severity differentiation [15], [20].
- **Quantification:** Measurement of the CEJ–alveolar bone crest distance and percentage bone loss. Only three studies address this: Jundaeng et al. achieved $mAP50=0.995$ [4]; Wimalasiri et al. reported $ICC=0.80$ [22]; and Tian et al. achieved $ICC=0.83$ [23].
- **Explainability:** Generation of interpretable visual outputs for clinical verification. Only 3 of 47 papers implement explainability (6.4%): Chen et al. [24] and Bilal et al. [10] use Grad-CAM, and Chen et al. [14] show feature map attention.
- **Clinical Integration:** Alignment with diagnostic workflows, reporting standards, and regulatory requirements. Only Schwendicke et al. [25] model economic impact, and only Lee et al. [26] measure clinician acceptance through a 65-dentist usability survey.

This six-component formulation stands in direct contrast to existing approaches that address each task in isolation – a fragmentation independently identified as critical by multiple systematic reviews [2], [6], [11], [12], [27], [28].

3.2 System Taxonomy

Existing approaches can be categorized into four distinct system archetypes based on task integration and diagnostic

scope. This taxonomy reveals both progression toward integration and persistent limitations preventing full clinical utility. Table I summarizes capability coverage across taxonomy dimensions.

1) *Single-Task Approaches:* Single-task systems address one diagnostic objective in isolation – caries detection, tooth segmentation, or periodontal classification – without integration. This accounts for the majority of reviewed studies (18 C1, 15 C2 papers). While several achieve high performance – CariesXrays FPCL reaching $AP50=84.1\%$ [14], Attention U-Net achieving $Recall=0.96$ [16], U-Net variants reporting $Dice=96.77\%$ [29] – none produce severity grading, periodontal assessment, or clinical explanations. The fundamental limitation is not performance but scope: optimizing one component without modeling interdependence with other diagnostic tasks produces clinically incomplete systems. Systematic reviews confirm this [6], [12], identifying absence of multi-task integration as the primary barrier to clinical translation.

2) *Multi-Task Approaches:* Seven reviewed papers adopt multi-task designs with varying completeness. Chen et al. [24] is the most clinically complete, classifying both periodontitis and caries on a per-tooth basis using EfficientNet-B0 with YOLOv7 localization ($ROC\ AUC=98.67\%$ for periodontitis, 98.31% for caries) with Grad-CAM explainability. However, this system operates exclusively on periapical radiographs – not panoramic OPGs – limiting direct applicability. Multi-task approaches expose a critical performance asymmetry: structural tasks (segmentation, enumeration) consistently achieve near-ceiling performance, while pathological tasks (disease detection, severity staging) remain substantially harder. Mureşanu et al. [7] quantified the real-world cost: YOLOv8 F1 dropped from 0.60 (internal) to 0.47 (external) – a 22% degradation from single-center training.

3) *Pipeline-Based Approaches:* Pipeline systems decompose diagnosis into sequential stages, reflecting the staged clinical reasoning dentists employ. Jundaeng et al. [4] implemented a precise periodontal pipeline: YOLOv8 Stage 1 for tooth segmentation ($mAP50=0.92$) feeds Stage 2 for CEJ and bone crest detection ($mAP50=0.995$, $Sensitivity=1.0$). Pornprasertsuk-Damrongsri et al. [16] combined YOLOv5s for tooth detection with Attention U-Net for caries segmentation ($Recall=0.96$, $F1=0.85$). The inherent limitation is error propagation: inaccuracies in early stages compound through subsequent stages. None of the reviewed pipelines address severity staging, and explainability appears in only one system [24] – operating on periapical, not panoramic, radiographs.

4) *End-to-End Approaches:* End-to-end models directly map radiographs to diagnostic outputs using deep architectures – primarily transformers. Bhoopalan et al. [5] applied Vision Transformer and Swin Transformer to 5,098 OPGs, achieving ViT accuracy=96% without spatial localization or severity grading. Van Nistelrooij et al. [30] combined Mask DINO with Swin-B Transformer, achieving tooth-level assessment performance ($FROC=0.988$) but with 40-second inference time unsuitable for clinical use. The fundamental barrier is data: the field lacks large-scale, multi-disease, multi-center

public datasets required for end-to-end learning.

The system-level taxonomy and capability gaps across different architectural paradigms are illustrated in Fig. 2

3.3 Task Hierarchy

Dental radiograph analysis follows a structured sequence of interdependent processing stages reflecting the actual cognitive process trained clinicians employ:

Early stages focus on anatomical structure identification, where systems perform well: tooth segmentation achieves pooled sensitivity = 0.92, specificity = 0.94 [28]; tooth enumeration reaches mAP50 = 0.978 [3]. Performance drops in the middle stages – disease detection and severity assessment. Detection on OPG reaches only mAP50 $\approx$ 0.61 [3], and severity staging appears in only two of 18 C1 papers [15], [20]. The terminal stages – explainability and structured reporting – are most neglected: explainability appears in only 3 of 47 papers (6.4%) [10], [14], [24], and structured report generation is absent from all systems. Dental AI is therefore a sequential reasoning problem, with early-stage failures propagating into downstream decisions. For systems targeting junior clinicians, the absence of visual explanations removes any educational value.

3.4 System-Level Limitations

Five system-level limitations persist across the reviewed literature, reflecting field-wide methodological choices:

- **Task Fragmentation:** No panoramic OPG system simultaneously addresses caries detection, severity staging, periodontal quantification, and explainability. Chen et al. [24] achieves dual-disease detection with explainability – closest to unified – but only on periapical radiographs.
- **Severity Staging Deficit:** Only Asadi [20] and Bizjak and Robic [15] provide caries severity differentiation among 18 C1 papers. For periodontal disease, the 2018 AAP/EFPP Stages I–IV [21] are not addressed by any system.
- **Explainability Absence:** Only 3 of 47 papers implement explainability (6.4%), contradicting field consensus [2], [3], [9], [35], [36]. The EU AI Act and FDA guidance require transparency for high-risk systems – requirements 43 of 47 systems cannot satisfy.
- **Dataset Fragmentation:** Approximately 60% of studies use private, single-center datasets causing heterogeneity values $I^2 = 88\text{--}96\%$ [12], [27], [28]. Mureşanu et al. [7] documented 22% F1 degradation from internal to external data. No large-scale, multi-disease, multi-center, publicly available panoramic dataset with standardized annotations exists.
- **Clinical Workflow Disconnection:** Except for Schwenicke et al. [25] (cost-effectiveness analysis) and Lee et al. [26] (65-dentist usability study), no system evaluates AI within actual clinical workflows. Most produce bounding boxes without structured reports or treatment recommendations.

4. BACKGROUND

4.1 Panoramic Radiography in Clinical Dentistry

Orthopantomography (OPG) produces a two-dimensional tomographic representation of the curved dental arches on a flat image plane. Where periapical or bitewing radiographs cover only limited regions, a single panoramic image captures all 32 permanent teeth, maxilla and mandible, and surrounding structures, making it the most information-dense single radiographic acquisition in routine dental practice. For caries detection, panoramic radiographs are less sensitive than bitewing for early proximal lesions [15], [16], [33]. For periodontal bone loss assessment, OPG is the standard screening tool, enabling visualization of the alveolar crest relative to the cemento-enamel junction (CEJ) across all teeth [2], [4], [27]. This modality distinction directly shapes the architectural requirements for AI systems addressing each task.

4.2 Deep Learning Architectures

The architectures reviewed fall into four functional families. First, CNNs with transfer learning – ResNet [32], DenseNet [17], [17], [19], VGG [18], EfficientNet [24], Inception [34], [35] – are used primarily for classification, leveraging ImageNet pre-training to compensate for limited dental dataset sizes. Second, YOLO-family one-stage object detectors (YOLOv5–YOLOv9) [3], [7], [13], [15], [16] dominate detection tasks for their real-time speed and strong bounding box localization. Third, U-Net and nested variants [6], [31], [37]–[39] are the standard for pixel-level segmentation, using skip connections to preserve spatial detail. Fourth, transformer architectures – ViT [5], Swin Transformer [30], [40], RT-DETR [41] – represent the most recent frontier, offering global attention at higher computational cost.

4.3 Evaluation Metrics

A persistent challenge is metric inconsistency, which prevents direct cross-study comparison [2], [6], [27], [28]. Object detection studies report mAP at IoU thresholds of 0.50 and 0.50:0.95. Segmentation studies use IoU and Dice coefficient. Classification studies report accuracy, AUC, sensitivity, specificity, and F1. Some studies [10], [24], [34] also report PPV and NPV, which have direct clinical relevance that aggregate accuracy scores can obscure. The absence of a standardized dental AI evaluation protocol is one of the most frequently cited methodological gaps across the reviewed corpus [1], [2], [27].

5. CARIES DETECTION AND SEVERITY CLASSIFICATION (C1)

Caries detection is the most extensively studied task in dental AI, accounting for 18 of the 47 reviewed papers. The studies span object detection, segmentation, classification, and systematic review methodologies across panoramic, bitewing, periapical, and intraoral RGB modalities.

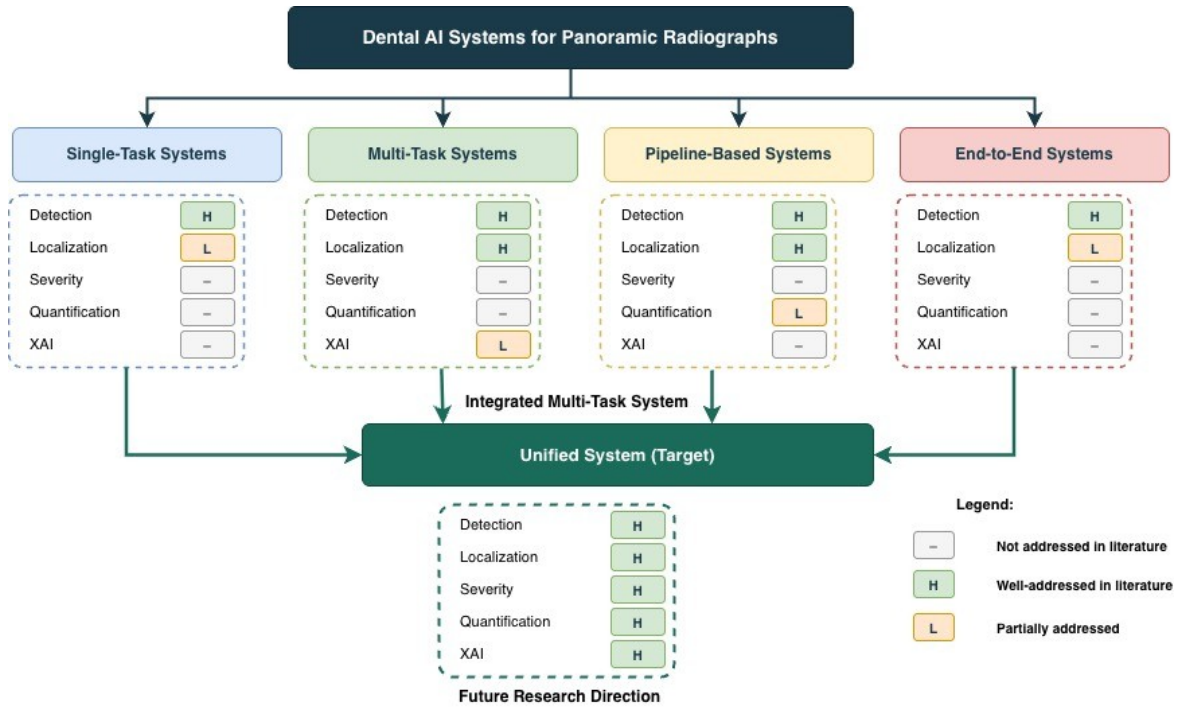


Fig. 2. System-level taxonomy of dental AI approaches for panoramic radiographs, illustrating four architectural paradigms (Single-Task, Multi-Task, Pipeline-Based, and End-to-End Systems) and their respective capability coverage across detection, localization, severity, quantification, and explainability (XAI) dimensions, alongside the proposed Unified System as a future research direction.

TABLE I
SYSTEM TAXONOMY — REPRESENTATIVE STUDIES BY DESIGN PATTERN AND CAPABILITY COVERAGE

System Type	Representative Studies	Detect	Localize	Severity	Quant.	XAI	Multi-Disease
Single-Task	[13], [14], [19], [31]–[34]	✓	Partial	×	×	×	×
Multi-Task	[3], [5], [7], [24], [30], [35]	✓	✓	×	×	Partial	Partial
Pipeline-Based	[4], [14]–[16]	✓	✓	×	Partial	×	×
End-to-End	[5], [30]	✓	Partial	×	×	×	Partial
Unified (Required)	<i>None in reviewed corpus</i>	✓	✓	✓	✓	✓	✓

✓ = fully addressed; × = not addressed; Partial = partially addressed.

Quant. = quantitative measurement; XAI = explainable AI outputs.

5.1 YOLO-Based Detection on Panoramic Radiographs

Ayhan et al. [13] proposed a two-stage pipeline targeting caries beneath fixed dental prostheses (FDPs), a scenario where metal artifacts from crowns and bridges actively obscure lesion visibility. YOLOv7 detected FDP regions (mAP = 0.968, F1 = 0.956) in Stage 1; a CBAM-enhanced YOLOv7 applied to cropped ROIs then achieved caries detection with mAP = 0.846 and F1 = 0.830 in Stage 2. The CBAM attention module produced measurable gains on low-contrast lesion boundaries beneath metallic artifacts.

Bizjak and Robic [15] applied a two-step YOLOv8 pipeline on the public DENTEX dataset, achieving tooth enumeration mAP@0.5 = 0.95 and lesion sensitivity = 0.70 (internal), 0.67 (external). That external evaluation is among the few genuine cross-site validations in the corpus and confirms the per-

formance drop that single-center training typically produces. Asadi [20] applied Faster R-CNN to the UFBA-UESC public dataset, achieving 94% overall accuracy but with a notable clinical weakness: recall for mild caries was 0.64 versus 0.69 for severe caries – the inverse of clinical need, where early lesions matter most. Pornprasertsuk-Damrongsri et al. [16] combined YOLOv5s for tooth detection with Attention U-Net for caries segmentation on 500 panoramic radiographs with bitewing-confirmed ground truth, achieving Recall = 0.96, F1 = 0.85 with Bland-Altman agreement analysis. An example of multi-tooth carious lesion detection with severity differentiation in panoramic radiographs is shown in Fig. 4.

verified bounding boxes) alongside the FPCL architecture, achieving AP=48.2%, AP50=84.1%, outperforming YOLOv8 (AP50=80.7%). CariesXrays is the most substantial public benchmark contribution in the C1 category.

5.5 Bitewing Caries and Cost-Effectiveness

Perez de Frutos et al. [33] evaluated RetinaNet, YOLOv5-M, and EfficientDet on 13,887 bitewing X-rays from the HUNT4 Oral Health Study, annotated by six dentists with a 197-image consensus test set. YOLOv5-M achieved mAP=0.647, F1=0.548, matching or surpassing individual expert annotators – evidence that deep learning can reach human expert performance on proximal caries detection. Schwendicke et al. [25] combined U-Net performance evaluation (accuracy=0.80, sensitivity=0.75 vs. dentist sensitivity=0.36) with a Markov lifetime cost-effectiveness simulation on 3,686 bitewing radiographs. The analysis projected increased tooth retention (64 vs. 62 years) and reduced cost per tooth (€298 vs. €322), yielding ICER=−13.9 €/year, which places AI caries detection in the cost-dominant category.

6. PERIODONTAL DISEASE AND ALVEOLAR BONE LOSS DETECTION (C2)

Periodontal disease detection from radiographs requires quantifying the spatial relationship between anatomical landmarks – specifically the distance between the cemento-enamel junction (CEJ) and the alveolar bone crest – across all teeth in the panoramic image. This task demands precise spatial localization and continuous measurement rather than binary classification. Fifteen papers in the C2 category reflect a less mature but rapidly advancing subfield.

6.1 Clinical Foundation

Chmielewski et al. [21] provide essential clinical context: the 2018 AAP/EFP classification defines periodontitis stages I–IV and grades A–C based on bone loss percentage, probing depth, tooth loss, and risk factors. Across 11 clinical studies covering 459 cases and 852 examiners, staging accuracy peaked at 90% but inter-examiner agreement remained variable. This is precisely the subjectivity problem that AI systems could address – provided they produce quantitative, staged outputs aligned with the 2018 criteria, rather than a binary bone loss label. Fig. 5 illustrates the spatial nature of periodontal bone loss assessment, which requires accurate identification of landmarks such as the CEJ and alveolar bone crest.

6.2 Landmark Detection and Bone Loss Quantification

Jundaeng et al. [4] reported the most technically complete periodontal AI system in the reviewed corpus. Their dual YOLOv8 pipeline on 2,000 panoramic radiographs achieved tooth segmentation (Accuracy=0.97, mAP50=0.92) and CEJ/bone level detection (Accuracy=0.98, Sensitivity=1.0, Specificity=0.98, F1=0.90, mAP50=0.995). A sensitivity of 1.0 for bone level detection carries practical weight: missed bone loss in a screening context is more harmful

than a false positive. Limitations include single-center data, images of uniformly high quality, and no caries integration or explainability.

Iacob et al. [27] synthesized evidence from 20 diagnostic accuracy studies covering 141,166 radiographs through PRISMA/QUADAS-2. U-Net, YOLO, ResNet, ConvNeXt, and DeNTNet have each been applied, but heterogeneity across studies prevents reliable pooling. Khubrani et al. [2] meta-analyzed 30 studies using APPRAISE-AI, reporting pooled sensitivity=87%, specificity=76%, accuracy=84% – figures that are useful as baselines but insufficient for autonomous clinical use.

Mathur et al. [18] applied VGG-16 to 1,874 panoramic radiographs for binary bone loss classification (accuracy=89.27%, sensitivity=83.27%). This approach, the most common in C2, is also the most clinically limited: it produces no staging information. Cerda Mardini et al. [42] attempted quantitative staging on 500 panoramic radiographs, reporting F1=0.29 for light-to-moderate bone loss and F1=0 for severe bone loss. The results honestly document how difficult quantitative staging is, and the model's failure on the most clinically critical severity class is worth noting. Kong et al. [39] integrated SRGAN super-resolution preprocessing with U-Net for bone loss segmentation (Dice=0.91, IoU=84.9%), the strongest segmentation metrics in C2. The results suggest that preprocessing quality directly affects segmentation accuracy, though the 698-image dataset is small and no disease staging is included.

Wimalasiri et al. [22] applied YOLOv8x and Keypoint R-CNN to 1,000 intraoral periapical X-rays, achieving ICC=0.80 for severity and 87% for bone loss pattern classification (horizontal vs. vertical). Pattern classification is absent from all other reviewed systems and has direct prognostic value for surgical planning. Tian et al. [23] developed the PANPOL mathematical model validated on 290 panoramic patients, achieving ICC=0.83, though this is a rule-based rather than a deep learning approach.

6.3 Periodontitis Classification

Bilal et al. [10] proposed PDCNET – a custom five-block CNN with Grad-CAM explainability – and evaluated it against EfficientNet-B0, DenseNet-201, VGG-16, VGG-19, Inception-V3, and MobileNet on a Kaggle dataset augmented with SMOTE-Tomek, achieving accuracy=98.39%, AUC=99.79%, F1=98.31%. These are the highest classification metrics in C2, though the original sample of 50 positive and 50 negative cases raises serious concerns about statistical reliability. The paper is useful for demonstrating the feasibility of Grad-CAM in periodontal AI. Ferrara et al. [9] reviewed 13 periodontal deep learning studies via PRISMA/QUADAS-2, reporting accuracy ranges of 73%–98.6% and noting that explainable AI is absent from every system reviewed. Lee et al. [26] assessed AI alveolar bone loss measurement usability through a survey of 65 dentists on 550 bitewing images – one of the few clinical usability studies in the entire corpus.

TABLE II
SUMMARY OF REVIEWED STUDIES — CARIES DETECTION AND SEVERITY CLASSIFICATION (C1)

Ref	Model	Dataset Size	Best Metric	XAI	Key Gap
[13]	YOLOv7 + CBAM	1,004 OPG	mAP=0.846, F1=0.830	None	Single-center, no XAI
[32]	YOLOv8, ResNet-50, EfficientDet, MobileNetV2	7,447 RGB	mAP@0.50=0.670	None	RGB only, no radio-graphs
[31]	U-Net, U-Net++, U-Net3+	1,500 OPG	IoU=0.67, Dice=0.60	None	Class imbalance, no severity
[37]	U-Net	6,075 OPG (pediatric)	F1=0.8675, AUC=0.76	None	Below 90% clinical threshold
[12]	Systematic Review (CNNs)	20 studies	Acc up to 0.98, AUC 0.98	None	High heterogeneity
[17]	DenseNet201	29,815 X-rays	Acc=91%, Cavity F1=0.27	None	Severe class imbalance
[14]	FPCL (Faster R-CNN + D2D-FPN + Contrastive)	6,000 OPG	AP=48.2%, AP50=84.1%	Attn maps	No severity/periodontal
[33]	YOLOv5-M, RetinaNet, EfficientDet	13,887 bitewing	mAP=0.647, F1=0.548	None	Bitewing only, no peri-odontal
[36]	Review: U-Net, ResNet, YOLOv7/v8, ensemble	Multiple studies	Acc 85–99%, F1 87–94%	Discussed	No original experiment
[25]	U-Net	3,686 bitewing	Acc=0.80, ICER=−€13.9/yr	None	Single-center, German context
[1]	Meta-analysis (CNNs)	21 studies, 17,190 images	Best Acc=98.8%	None	I^2 =88–96% heterogeneity
[29]	UNet, UNet+, UNet++	500 images	Acc=98.03%, Dice=96.77%	None	Small dataset, no clinical val.
[34]	InceptionResNet-v2	935 OPG / 7,304 teeth	Acc=0.94, F1=0.94	None	No localization, no peri-odontal
[38]	Faster R-CNN + U-Net (EfficientNet-B0)	597 OPG	IoU=0.608, Dice=0.743	None	No severity, no periodontal
[6]	Systematic Review (U-Net, Mask R-CNN, ResNet)	42 studies	Various (review)	None	No severity differentiation
[15]	YOLOv8 two-step pipeline	1,005 OPG + 85 external	mAP@0.5=0.95, Sens=0.70	None	Caries only, no severity
[20]	Faster R-CNN	1,000 OPG	Acc=94%, recall=0.64	Mild	Weak mild caries detection
[16]	YOLOv5s + Attention U-Net	500 OPG, 14,997 teeth	Recall=0.96, F1=0.85	None	Single-disease, no peri-odontal
[19]	DenseNet-121	14,498 tooth images	Acc=93.17%, F1=87.49%	None	No localization, no severity

Kim et al. [43] proposed an unsupervised few-shot learning architecture combining UNet, CVAE, and clustering methods (k-means/DBSCAN/GMM) on 1,372 panoramic radiographs. The approach achieved a 14% improvement over supervised ViT, suggesting that label-efficient learning may be a practical path forward in data-scarce dental AI settings. Karamuftuoğlu et al. [41] benchmarked YOLOv12x, RT-DETR-L, and RT-DETR-X on a 387-image pediatric panoramic dataset for furcation lesion detection, achieving mAP@0.5=0.434. The modest result reflects both the difficulty of the task and the constraints of small single-center datasets.

7. MULTI-TASK AND END-TO-END OPG DIAGNOSIS (C3)

Seven papers address multiple diagnostic tasks within a single pipeline applied to panoramic radiographs – the architectural direction most consistent with real clinical utility.

Chen et al. [24] report the most clinically complete dual-disease system in the corpus, combining YOLOv7 for tooth localization with EfficientNet-B0 for simultaneous multi-label classification of periodontitis and dental caries on periapical X-rays, with Grad-CAM visualization. EfficientNet-B0 achieved

ROC AUC = 98.67% for periodontitis and 98.31% for dental caries with 10-fold cross-validation. The critical limitation is modality: periapical, not panoramic.

Nassiri and Akhloufi [3] compared YOLO v5–v9 on DENTEX across quadrant segmentation, tooth enumeration, and disease detection. YOLOv9e-seg achieved mAP50=0.978 for tooth enumeration and 0.959 for segmentation, while pathology detection peaked at mAP50=0.61 – confirming that structural tasks are tractable while pathology detection remains the harder problem.

Van Nistelrooij et al. [30] combined OdontoAI and DENTEX datasets with Mask DINO (ViT backbone) and Swin-B Transformer, achieving FROC=0.988 for tooth segmentation and AUC=0.996 for impacted teeth, the strongest tooth-level assessment results reviewed. A 40-second inference time per image is the primary barrier to clinical use. Gamal et al. [35] demonstrated that a weighted average ensemble of CNN architectures (Custom CNN, ResNet-50, InceptionV3, DenseNet121, VGG19) achieves accuracy = 92.93% on 1,814 cropped tooth images. ESRGAN super-resolution preprocess-

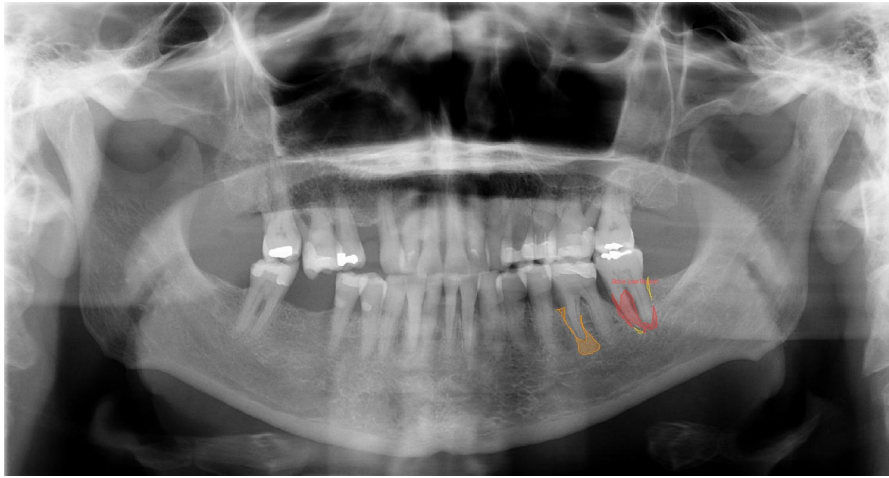


Fig. 5. Example panoramic radiograph illustrating periodontal bone loss regions. Accurate assessment requires spatial measurement between the cemento-enamel junction and alveolar bone crest.

ing contributed roughly 3% of that accuracy gain.

Mureșanu et al. [7] provided the most methodologically transparent external validation in C3: YOLOv8 tested on a Romanian clinical dataset showed internal $F1=0.60$ degrading to external $F1=0.47$ – a 22% drop that directly measures the generalization cost of single-center training. Bhoopalan et al. [5] applied ViT and Swin Transformer to 5,098 OPG images, achieving ViT accuracy=96%, precision=95.8%, recall=96.2%. These are strong results for image-level classification, though the system provides neither localization nor segmentation. Zhou et al. [11] reviewed datasets, methodologies, and emerging challenges in dental image analysis, reinforcing the need for standardization across the field.

8. EXPLAINABLE AI IN DENTAL DIAGNOSTICS (C4)

Of the 47 papers reviewed, only three report any explainability output: Chen et al. [24] use Grad-CAM for dual-disease classification on periapical X-rays; Bilal et al. [10] apply Grad-CAM to periodontal disease classification; and Chen et al. [14] include attention visualization in feature maps for caries detection. This constitutes **6.4%** of the reviewed corpus – a figure that directly quantifies the explainability deficit in dental AI.

Gradient-weighted Class Activation Mapping (Grad-CAM) generates a coarse localization map of the image regions most influential for a CNN prediction. In dental diagnostics, it serves three practical functions: (i) it allows clinicians to verify that the model attends to anatomically plausible regions; (ii) it can function as a visual teaching tool for junior clinicians learning radiographic interpretation; and (iii) it creates an audit trail relevant to the EU AI Act and FDA guidance on AI/ML-based medical devices, both of which classify AI diagnostic tools as high-risk systems requiring transparency and human oversight [24].

Field consensus holds that explainability is necessary for clinical adoption [3], [9], [35], [36], yet the implementation rate is 6.4%. For systems targeting junior clinicians and dental

students, this is more than an oversight – a visual justification is what transforms AI-assisted diagnosis from prediction into education. SHAP, LIME, and transformer attention visualization remain entirely unused in dental AI; transformer attention maps are already intrinsic to the architecture and add no computational overhead [5].

9. COMPARATIVE ANALYSIS OF REPRESENTATIVE METHODS

10. DATASETS, BENCHMARKS, AND CLINICAL VALIDATION (C5)

Dataset availability and quality are the most frequently cited limitations across all 47 reviewed papers. The key public benchmarks in the field are as follows:

- **DENTEX**: 1,005 panoramic radiographs with quadrant, tooth, and pathology annotations from three institutions [3], [15], [30].
- **CariesXrays**: 6,000 hospital-grade panoramic images with 13,783 caries bounding boxes [14].
- **Tufts Dental Database**: Approximately 1,000 panoramic radiographs with tooth segmentation masks [40], [42]–[44].
- **HUNT4 Oral Health Study**: 13,887 bitewing X-rays with six-dentist annotations and a 197-image consensus set [33].
- **UFBA-UESC**: 4,000 panoramic images [20].

Despite these contributions, roughly 60% of reviewed primary studies rely on private, single-center institutional data. Systematic reviews [1], [2], [12], [27], [28] consistently report I^2 heterogeneity values of 88–96%, which is the primary barrier to reliable evidence synthesis. The performance degradation from internal to external validation documented by Mureșanu et al. [7] – $F1$ dropping from 0.60 to 0.47 – directly measures the cost of single-center training.

Waqas et al. [45] addressed data scarcity through GAN-based synthetic OPG generation (DCGAN, WGAN) on 4,777

TABLE III
SUMMARY OF REVIEWED STUDIES — PERIODONTAL DISEASE AND BONE LOSS (C2)

Ref	Model	Dataset Size	Best Metric	XAI	Key Gap
[4]	Dual YOLOv8 (teeth + CEJ/bone)	2,000 OPG	mAP50=0.995, Sens=1.0	None	Single-center, no caries
[21]	Clinical review (no ML)	459 cases, 852 examiners	Stage acc=90%	None	No AI system proposed
[27]	Systematic Review (CNN, YOLO, U-Net)	141,166 radiographs	Various (review)	None	High heterogeneity
[18]	VGG-16 (transfer learning)	1,874 OPG	Acc=89.27%	None	Binary only, no staging
[2]	Meta-analysis (CNN, YOLO, U-Net, SVM)	30 studies	Sens=87%, Spec=76%	None	Pooled heterogeneity
[23]	PAN-POL mathematical model	290 PAN patients	ICC=0.83	None	Not deep learning, manual
[26]	Faster R-CNN + DeepLab v3+	550 bitewing	Acc=82–87% (severe)	None	Bitewing only, no OPG
[42]	Hybrid DCNN (Xception) + rule-based PBL	500 OPG	F1=0.29 (lt-mod), F1=0 (severe)	None	Fails on severe PBL
[22]	YOLOv8x + Keypoint R-CNN	1,000 IOPA	ICC=0.80, Pattern Acc=87%	None	Periapical only, no OPG
[39]	SRGAN + U-Net	698 OPG	Dice=0.91, IoU=84.9%	None	No severity, small dataset
[9]	Systematic Review (CNNs, hybrid)	13 studies	Acc 73–98.6%	None	No XAI in any reviewed study
[10]	PDCNET (custom CNN) + Grad-CAM	50+50 orig (SMOTE aug.)	Acc=98.39%, AUC=99.79%	Grad-CAM	Dataset too small
[41]	YOLOv12x, RT-DETR-L, RT-DETR-X	387 OPG (pediatric)	mAP@0.5=0.434	None	Small dataset, modest mAP
[43]	UNet + CVAE + k-means/DBSCAN/GMM	1,372 OPG	+14% over supervised ViT	None	No bounding boxes

real images. Expert assessors rated synthetic radiographs as realistic above 50% of the time relative to real images, though downstream gains in disease detection accuracy were not demonstrated. Ameli et al. [46] applied YOLOv8 to 1,000 OPGs for automated quality assessment (contrast/density accuracy = 97.9%, artifact accuracy = 87.2%, overall quality = 79.3%), establishing automated quality filtering as a viable preprocessing gate for diagnostic pipelines. Delamare et al. [8] documented heterogeneous quality criteria across 41 panoramic ML studies and argued that standardized quality protocols are a prerequisite for reproducible AI development.

11. TOOTH SEGMENTATION AND NUMBERING (C6)

Four papers address tooth detection, segmentation, and numbering – tasks that are prerequisites for tooth-specific disease attribution. FDI-standard tooth numbering is needed to report that caries is in tooth 26 rather than somewhere in the panoramic image.

Bonfanti-Gris et al. [28] synthesized 20 tooth detection and segmentation studies in a PRISMA/QUADAS-2 meta-analysis, reporting pooled sensitivity = 0.92 and specificity = 0.94 with DOR = 186 – strong aggregate evidence, though $I^2 \approx 96\%$ confirms severe heterogeneity. Zhang et al. [40] proposed a multi-stage system (MobileNetV3 + SADP-Net Mask R-CNN + Swin Transformer + YOLOv5) across three public datasets,

achieving AP50 = 97.3% on Tufts and 98.1% on O2PR for tooth instance segmentation. Yaswanth et al. [44] applied Attention-integrated U-Net++ to the Tufts dataset, achieving mAP = 96.4% and Dice = 90%. Adnan and Umer [47] contributed the Orthopantomogram Teeth Segmentation and Numbering Dataset – 250 expert-annotated OPGs on Zenodo – while explicitly noting that disease annotations are absent and that expansion to caries and periodontal labeling is needed.

12. CROSS-STUDY TRENDS AND OBSERVATIONS

This section presents a synthesized analysis of patterns that recur across the 47 reviewed studies. Drawing on the global comparison in Table IV, several trends emerge across model design, dataset characteristics, task complexity, and deployment challenges.

12.1 Model Architecture Trends

1) Transformer-Based vs. CNN-Based Architectures: Transformer-based architectures such as ViT and Swin Transformer capture global contextual information well, which benefits classification and multi-label diagnosis [30]. In practice, however, CNN-based architectures dominate the reviewed literature, particularly in detection and pipeline-based systems [17]. Their efficiency, lower computational requirements, and reliable behavior on moderately sized datasets explain this dominance.

TABLE IV
COMPARISON OF REPRESENTATIVE DEEP LEARNING METHODS FOR DENTAL RADIOGRAPH ANALYSIS. PERFORMANCE METRICS ARE REPORTED AS PROVIDED IN THE RESPECTIVE STUDIES AND ARE NOT DIRECTLY COMPARABLE DUE TO VARIATIONS IN TASKS, DATASETS, AND EVALUATION PROTOCOLS.

Ref	Task	Scope	Dataset (Size)	Modality	Model	Best Metric	XAI	Clinical Val.	Key Limitation
[13]	Caries Detection	Single-task	Private OPG (1,004)	OPG	YOLOv7 + CBAM	mAP=0.846	No	Internal	No external validation
[4]	Bone Loss + Periodontal	Multi-task	Private OPG (2,000)	OPG	YOLOv8	mAP50=0.995	No	Internal	Single-center dataset
[24]	Multi-disease Detection	Multi-task	Periapical (11k)	X-ray	YOLOv7 + EfficientNet	AUC≈98%	Yes	Internal	Not panoramic
[31]	Caries Segmentation	Single-task	DNS (1,500)	OPG	U-Net3+	Dice=0.60	No	Internal	Small dataset
[37]	Caries Segmentation	Single-task	Pediatric OPG (6,075)	OPG	U-Net	F1=0.867	No	Internal	No severity staging
[35]	Multi-disease Classification	Multi-task	Roboflow (1,075)	OPG	CNN Ensemble	Acc=92.9%	No	Internal	No explainability
[14]	Caries Detection	Single-task	CariesXrays (6,000)	OPG	Faster R-CNN	AP50=84.1%	Partial	Internal	Single-disease
[33]	Caries Detection	Single-task	HUNT4 (13k)	Bitewing	YOLOv5	mAP=0.647	No	External	Not panoramic
[3]	Multi-task OPG Analysis	Multi-task	DENTEX (~1,000)	OPG	YOLOv9-seg	mAP=0.978	No	Internal	Limited pathology
[30]	Tooth + Diagnosis	Multi-task	OdontoAI (1,661) + DENTEX (682)	OPG	Mask DINO + Swin	mAP=0.848	No	Internal	Slow pipeline
[34]	Tooth + Caries	Multi-task	935 PAN + 18k teeth	OPG	InceptionResNet	Acc=0.998	No	Internal	No localization
[38]	Caries Segmentation	Single-task	597 PAN	OPG	Faster R-CNN + U-Net	Dice=0.743	No	Internal	Single-disease
[15]	Caries Detection	Single-task	DENTEX + external	OPG	YOLOv8	mAP=0.95	No	External	No segmentation
[20]	Caries Severity	Single-task	1,000 PAN	OPG	Faster R-CNN	Acc=94%	No	Internal	Weak mild detection
[16]	Caries Detection	Multi-stage	500 PAN	OPG	YOLOv5 + U-Net	Recall=0.96	No	Internal	Single-disease
[18]	Bone Loss Classification	Single-task	1,874 OPG	OPG	VGG-16	Acc=89.27%	No	Internal	No localization
[26]	Bone Loss Detection	Single-task	Bitewing (550)	Bitewing	Faster R-CNN + DeepLab	Acc≈85%	No	Internal	Limited modality
[22]	Bone Loss Severity	Single-task	DenPAR (1,000)	IOPA	YOLOv8 + KRCNN	ICC=0.80	No	Internal	Not panoramic
[39]	Bone Loss Segmentation	Single-task	CBNUH (100) + Kaggle (~600)	OPG	SRGAN + U-Net	Dice=0.91	No	Internal	No detection
[7]	Multi-disease Detection	Single-task	Multi-center panoramic	OPG	YOLOv8	F1=0.60→0.47	No	External	Poor generalization
[40]	Tooth Segmentation	Multi-stage	Tufts + O2PR	OPG	Mask R-CNN + YOLO	mAP=77.4	No	Internal	No disease labels
[28]	Tooth Segmentation Review	Review	20 studies	Multi	CNNs	Sens=0.92	No	Review	High heterogeneity
[2]	Bone Loss Review	Review	30 studies	Multi	CNNs	Acc=84%	No	Review	No standardization
[11]	Deep Learning Survey	Review	260 studies	Multi	Various	—	No	Review	No unified system
[10]	Periodontal Classification	Single-task	Kaggle (~1,000)	X-ray	PDCNET	Acc=98.39%	Yes	Internal	No localization

Transformer-based models require large annotated datasets and exhibit higher inference latency, which reduces their practicality in most clinical environments [30]. CNN-based approaches therefore remain more suitable for deployment under current dataset constraints, despite their representational limitations.

2) *Dominance of YOLO in Detection Tasks*: YOLO variants are the most widely used models for caries localization, tooth detection, and bone loss identification across the reviewed literature [3], as reflected in Table IV. Single-pass inference with near real-time performance makes them well suited to clinical applications, where latency matters.

3) *Ensemble Methods*: Ensemble methods can improve classification robustness by aggregating predictions from multiple models, but the associated increase in computational cost

limits their practicality in real-time clinical settings [35].

12.2 Data and Generalization

Dataset size improves model performance but does not guarantee generalization. Large datasets can still suffer from class imbalance, depressing task-specific metrics [17]. More consequentially, models trained on single-center data consistently underperform when tested on external data.

This generalization gap is a structural issue rather than an idiosyncrasy of individual studies. Models trained on panoramic data from a single institution degrade under multi-center conditions [7], and systematic reviews confirm high heterogeneity attributable to differences in imaging protocols, annotation standards, and model architectures [2]. Dataset diversity, multi-

center representation, and standardized evaluation protocols are necessary conditions for clinically reliable systems.

12.3 Task Complexity and Learning Paradigm

1) *Structural Tasks vs. Disease Detection*: Structural tasks such as tooth segmentation and enumeration are relatively low-variance problems; performance is consistently high across architectures [28]. Disease detection – particularly early-stage caries – is substantially harder due to subtle visual features with high intra-class variability.

The implication is that anatomical recognition and pathological detection place fundamentally different demands on model design and data quality. Within the same framework, accuracy typically falls when transitioning from structural to disease detection tasks [3].

2) *Multi-Task Learning vs. Single-Task Models*: Most reviewed studies adopt single-task formulations; only a small subset attempts multi-task learning, as shown in Table IV. This is a practical limitation, since real-world dental diagnosis requires simultaneous evaluation of multiple conditions. Studies that do combine objectives within a single framework [24] remain rare and are often restricted in scope or modality.

12.4 Deployment and Evaluation Challenges

1) *Explainable AI*: Explainability appears in approximately 6.4% of reviewed studies, with the remaining systems operating as black-box models [24]. Without interpretable outputs, clinical trust is reduced and regulatory approval becomes difficult. Explainability is therefore a deployment requirement, not an optional feature.

2) *Lack of Standardized Evaluation Metrics*: Studies report heterogeneous measures – mAP, Dice, IoU, accuracy, F1 – with inconsistent thresholds and experimental setups, making cross-study comparison unreliable. As Table IV shows, this inconsistency complicates benchmarking across models and tasks. Systematic reviews confirm that metric variability is a primary driver of inter-study heterogeneity [28]. Standardized evaluation protocols are needed before the field can make reliable comparative progress.

12.5 Final Observation

CNN-based and YOLO-driven approaches dominate current dental AI, while transformer-based models are emerging but remain resource-intensive. Structural tasks are solved reliably; disease detection – especially early-stage and quantitative assessment – is not. The gap between reported performance on individual benchmarks and the requirements of a clinically deployable pipeline is large, and closing it requires integrated, explainable, generalizable systems rather than continued refinement of isolated sub-tasks.

13. CRITICAL ANALYSIS OF EXISTING METHODS

13.1 Metrics and Evaluation Limitations

The reviewed literature lacks standardized evaluation practices. Studies report mAP, Dice coefficient, accuracy, AUC,

and F1-score with varying thresholds and experimental setups, making direct comparison across methods unreliable and obscuring which approaches are genuinely superior [2], [27], [28].

Many studies report strong results under controlled conditions but do not use benchmarking protocols that would permit cross-study comparison. Only a small subset of works adopt consistent evaluation settings. Without unified benchmarks and evaluation standards, the reproducibility and reliability of published results remain fundamentally constrained.

13.2 Dataset Bias and Generalization Issues

A large proportion of reviewed studies use data from a single institution or imaging setup. Single-center datasets fail to capture variability across populations, imaging devices, and acquisition conditions [1], [12], and models trained on them consistently degrade when evaluated externally, as multiple studies confirm [7].

Public datasets such as DENTEX and CariesXrays provide useful baselines but do not cover multi-disease, multi-center clinical variability [3], [14]. Without diverse, standardized, multi-institutional datasets and consistent validation protocols, the generalizability of current systems remains limited.

13.3 Clinical Integration Gap

Most systems are evaluated offline and produce intermediate outputs – bounding boxes, segmentation masks, classification labels – without connecting to real-time clinical decision support. Clinician involvement in model development and evaluation is limited; only a small number of studies assess real-world diagnostic impact in terms of accuracy, treatment planning, or workflow efficiency [25], [26].

The absence of explainability compounds this problem. Black-box systems without interpretable output create a practical disconnect between technical performance and clinical applicability. Most published systems are not aligned with the requirements of routine clinical use.

13.4 Fragmentation of Existing Approaches

As Table I shows, the majority of reviewed studies target a single task – detection, segmentation, or classification – with only a small fraction exploring integrated multi-task frameworks. Real-world dental diagnosis requires simultaneous evaluation of caries, periodontal status, severity staging, and quantitative measurement. Existing systems do not jointly address these components and therefore cannot reflect the full complexity of a clinical workflow.

No reviewed system operating on panoramic radiographs simultaneously integrates detection, localization, severity staging, quantification, and explainability within a unified framework. This is a structural limitation of the field, not a property of individual models.

Progress will require a shift from task-specific optimization toward system-level design, where diagnostic components are jointly modeled and aligned with clinical requirements.

These limitations collectively highlight the need for unified, clinically aligned, and generalizable dental AI systems, which are discussed in the following section.

14. CROSS-CUTTING GAP ANALYSIS

Synthesizing findings across all 47 reviewed papers, five interconnected gaps emerge that consistently constrain clinical translation.

14.1 Gap 1: Single-Disease Focus

Single-disease training is the dominant pattern across all categories. Of 47 papers reviewed, only Chen et al. [24] simultaneously detect both caries and periodontal disease – and exclusively on periapical, not panoramic, radiographs. On panoramic radiographs, no reviewed study provides simultaneous caries detection, severity staging, and periodontal bone loss quantification within a single pipeline. Patients routinely present with co-occurring conditions that must be assessed from a single exposure; no current system addresses this.

14.2 Gap 2: Absence of Automated Severity Staging

Of 18 C1 papers, only Asadi [20] and Bizjak and Robic [15] distinguish severity beyond binary caries presence. In C2, only Wimalasiri et al. [22] address bone loss severity patterns. The 2018 AAP/EFP classification [21] defines four periodontitis stages; none are addressed by any panoramic OPG system reviewed. Severity staging determines treatment planning, referral, and prognosis. Detection without staging yields limited clinical decision support.

14.3 Gap 3: Explainability at 6.4%

Only 3 of 47 papers implement any XAI output. Regulatory frameworks including the EU AI Act classify AI diagnostic tools as high-risk systems requiring transparency and human oversight – requirements that 43 of the 47 reviewed systems cannot satisfy. For junior clinicians, explainability is not optional; it is the mechanism through which AI-assisted diagnosis supports learning rather than replacing judgment. The consensus that explainability is necessary for clinical adoption [3], [9], [35], [36] is not reflected in implementation rates.

14.4 Gap 4: Dataset Fragmentation

Approximately 60% of reviewed primary studies use private, single-center datasets. Systematic reviews [1], [2], [12], [27], [28] consistently report $I^2 = 88\text{--}96\%$ heterogeneity as the primary barrier to evidence synthesis. The real cost is quantified by Mureşanu et al. [7]: F1 drops 22% from internal to external validation. The field needs a single large-scale, multi-disease, multi-center, publicly available panoramic OPG dataset with standardized caries and periodontal annotations. Nothing in the reviewed literature currently provides this.

14.5 Gap 5: Clinical Workflow Integration

With the exception of Schwendicke et al. [25] and Lee et al. [26], no reviewed study evaluates AI within an actual clinical workflow or measures clinician acceptance and diagnostic impact. Most systems produce bounding boxes or segmentation masks without clinical report generation, severity summaries, or treatment recommendations. The cost analysis by Schwendicke et al. [25] (ICER = -13.9 €/year for AI-assisted caries detection) is an isolated example of the real-world impact evaluation the field needs.

15. RESEARCH ROADMAP TOWARD UNIFIED AND CLINICALLY DEPLOYABLE DENTAL AI SYSTEMS

Building on the limitations identified in Section 13, this section outlines a structured roadmap toward unified, clinically deployable dental AI systems.

15.1 Short-Term Goals (1–2 Years)

Public Multi-Disease Panoramic Dataset: The most immediate bottleneck is the absence of a large-scale, standardized panoramic dataset with multi-disease annotations. A consortium-driven effort is needed to produce at least 5,000 panoramic radiographs with consistent annotations for caries (per-tooth, with severity), periodontal bone loss (CEJ–bone crest distance), and tooth numbering (FDI standard).

Datasets such as CariesXrays [14] and DENTEX [3] offer partial solutions but lack comprehensive multi-disease annotations. Extending and integrating them is a practical near-term step.

Explainability as a Standard Component: Explainability should be incorporated by default, using Grad-CAM or attention visualization, with validation against expert annotations to confirm anatomical relevance [10], [17].

15.2 Mid-Term Goals (3–5 Years)

Unified Multi-Disease System for Panoramic Radiographs: The most pressing research direction is a unified multi-disease system that simultaneously detects caries, performs severity staging, quantifies periodontal bone loss, and supports multi-condition diagnosis from a single panoramic radiograph.

This requires multi-head architectures with shared feature representations and task-specific outputs. Individual components exist in the literature, but their integration into a coherent framework remains unsolved [3], [4], [7], [24].

Clinical Alignment with Standardized Diagnostic Frameworks: Future systems should produce outputs that align with established classification standards – ICCMS for caries and the 2018 AAP/EFP classification for periodontal disease [21]. This would bridge the gap between algorithmic output and clinically actionable decisions.

15.3 Long-Term Goals (Clinical Deployment)

Prospective Clinical Validation and Deployment: The field must move beyond retrospective evaluation. Randomized studies assessing diagnostic accuracy, clinician efficiency, and

patient outcomes with and without AI assistance are needed, following standardized reporting protocols to allow comparisons across settings.

The cost-effectiveness framework of Schwendicke et al. [25] provides a foundation that should be extended to multi-disease systems and panoramic radiographs. Prospective validation of this kind is essential for both regulatory approval and clinical acceptance.

Final Perspective: This roadmap describes a path from data standardization and model explainability toward fully integrated, prospectively validated systems – a concrete progression for the next generation of dental AI research.

16. CONCLUSION

This survey reviewed 47 deep learning studies (2021–2026) on dental caries detection, periodontal bone loss assessment, and related tasks from panoramic and bitewing dental radiographs. Deep learning has reached clinically competitive performance on individual sub-tasks: caries detection achieves 96% recall with Attention U-Net [16]; periodontal landmark detection reaches mAP50=0.995 with YOLOv8 [4]; dual-disease classification attains AUC > 98% on periapical X-rays with EfficientNet-B0 and Grad-CAM [24]; and AI-assisted caries detection is cost-dominant relative to unaided clinical assessment, with ICER = −13.9 €/year [25].

Five structural gaps persist across the reviewed corpus: single-disease models that ignore clinical co-occurrence; absent severity staging despite its centrality to treatment planning; a 6.4% explainability rate despite consensus on its necessity; dataset fragmentation from private single-center collections; and minimal clinical workflow integration.

No existing system simultaneously addresses caries detection, severity staging, periodontal bone loss quantification, and explainability on panoramic radiographs within a single pipeline. This is the unmet need that the reviewed literature collectively establishes. The technical components exist – YOLO detection, U-Net segmentation, Grad-CAM explainability, EfficientNet classification, and transformer-based attention have each been demonstrated on dental radiographic data. The challenge is integration, clinical validation, and standardization. Systems that are accurate but not interpretable or clinically grounded cannot serve the junior clinicians and dental students who stand to benefit most from AI-assisted diagnostic support.

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